

Structural Rank, Decision Entropy, and Thermodynamic Selection in Finite Information-Processing Systems

Tristan Simas
McGill University
`tristan.simas@mail.mcgill.ca`

April 15, 2026

Abstract

Exact resolution in finite bounded physical systems carries irreducible thermodynamic cost. Under Landauer calibration, any exact-resolution cycle satisfies

$$E \geq k_B T H_{\text{nats}}(D),$$

where $H_{\text{nats}}(D)$ is the natural-log entropy of the decision quotient. In the canonical binary encoding studied here, this bound sharpens to

$$E \geq \text{DOF}(A) k_B T \ln 2,$$

and the rank-1 regime is the unique thermodynamic ground state: every system with more than one degree of freedom lies strictly above the minimum per-cycle resolution cost.

The England replication inequality,

$$\Delta S_{\min}(k) - \Delta S_{\min}(1) \geq k_B \ln k,$$

is also obtained in the same framework. The proof reduces the entropy premium of replication to finite counting: a k -coordinate system has 2^k states, and the elementary inequality $k \leq 2^{k-1}$ yields the gap.

These thermodynamic statements are derived from a finite structural theorem. Exact resolution in a bounded system occurs through finitely many discrete acquisition events. The associated canonical decision problem records those events as Boolean coordinate reads, and the number of independent coordinates is identified exactly with the structural rank of the encoded decision problem:

$$\text{DOF}(A) = \text{srnk}(\text{canonicalDP}(A)).$$

The decision quotient therefore has at most $2^{\text{DOF}(A)}$ optimal-action classes, so its entropy is controlled by the same coordinate count that governs exact physical resolution.

All theorems are machine-checked in Lean 4 with no `sorry` placeholders. The mechanized artifact records the proof provenance in full.

Keywords: thermodynamics, Landauer principle, decision entropy, structural rank, bounded physical systems, formal verification

1 Introduction

Finite bounded physical systems resolve decisions through finitely many discrete acquisition events on bounded horizons. Under Landauer calibration [10, 1], exact resolution therefore carries irreducible thermodynamic cost. The formal object is a bounded decision system, represented in Lean by `Architecture`, carrying a positive degree-of-freedom count together with a canonical binary decision encoding `canonicalDP`. This counting parameter is exactly the structural rank of the encoded decision problem, it bounds the entropy of the decision quotient, and it determines the minimum per-cycle resolution cost. The quotient object is closer to zero-error and confusability-based information than to average-case coding [19, 9, 11]. All claims are verified in Lean 4 [4] with Mathlib support [21] and zero `sorry` placeholders.

1.1 Central Result

The convergence theorem proved in Section 5 is:

$$\begin{array}{c}
 \underbrace{\text{DOF}(A) = 1}_{\text{one-coordinate regime}} \iff \underbrace{\text{srnk}(\text{canonicalDP}(A)) = 1}_{\text{structural rank}} \\
 \iff \underbrace{\text{tractable sufficiency for canonicalDP}(A)}_{\text{complexity}} \iff \underbrace{\text{minimum per-cycle thermodynamic cost}}_{\text{physics}}
 \end{array}$$

An imported coherence theorem gives a separate single-source reading of the same rank-1 point:

$$\text{SSOT}(A) \iff \text{DOF}(A) = 1,$$

where $\text{SSOT}(A)$ denotes the coherent single-source condition that one locus is authoritative, every remaining encoding is a derived view, and all reachable states remain coherent.

The structural-rank and thermodynamic clauses are central. The imported coherence statement is a companion interpretation of the same rank-1 regime, not a premise of the mathematical-physics chain.

1.2 Contributions

1. **Finite Physical Acquisition:** bounded regions admit finitely many acquisition events, those events are discrete state transitions, and exact resolution requires a sufficient coordinate set.
2. **DOF-Structural-Rank Identification:** the canonical decision problem attached to an n -degree-of-freedom system has structural rank n .
3. **Entropy-Rank Control:** the number of decision classes is at most $2^{\text{DOF}(A)}$, so the decision entropy is bounded by the degree-of-freedom count.
4. **Landauer-Linear Resolution Cost:** under Landauer calibration, exact-resolution cost is bounded below by $\text{DOF}(A) k_B T \ln 2$.
5. **Energy-Information Duality:** the same system satisfies $E \geq k_B T H_{\text{nats}}(D)$, linking thermodynamic cost directly to the entropy of the decision quotient.
6. **Finite-Budget No-Collapse:** bounded budget, positive per-bit cost, and exponential lower-bound growth jointly preclude physical collapse of the higher-rank regime.

7. **England Replication Inequality (Theorem 5.6):** $\Delta S_{\min}(k) - \Delta S_{\min}(1) \geq k_B \ln k$. The proof reduces the entropy gap to counting via $k \leq 2^{k-1}$.¹
8. **Rank-1 Convergence (Theorem 5.5):** the rank-1 regime is simultaneously the one-coordinate regime, the tractable sufficiency regime, and the thermodynamic ground state.

Physical significance. Once finite acquisition is fixed as the physical interface, the degree-of-freedom count becomes simultaneously an interaction dimension, an entropy bound, and a Landauer-calibrated cost coordinate. The result concerns exact resolution structure in matter.

1.3 Scope

The mathematical structure links structural rank, quotient entropy, complexity, and thermodynamic cost. Theorems are stated for bounded decision systems represented by the Lean object **Architecture** and their canonical decision encoding, not for arbitrary physical systems without mediation. Architectural and programming interpretations are downstream readings of the same formal core.

1.4 Organization

Section 2 defines the structural model, the finite-acquisition interface, and the canonical decision encoding. Section 3 derives the structural-rank and entropy consequences of that encoding. Section 4 derives the thermodynamic consequences. Section 5 proves the convergence theorem and the England inequality. Section 6 situates the paper relative to thermodynamics, information theory, and formalized complexity. Section A describes the Lean mechanization.

2 Foundations

The formal objects that carry the mathematical-physics content are a positive degree-of-freedom count, a canonical binary decision encoding, structural rank, and decision entropy.

2.1 Formal Object

Definition 2.1 (Bounded Decision System). A *bounded decision system* is a finite bounded physical system equipped with a positive integer $\text{DOF}(A)$. In the mechanized artifact the corresponding Lean object is named **Architecture**, but the results used below depend only on the degree-of-freedom count and on the canonical decision encoding.

Interpretation. $\text{DOF}(A)$ counts independent coordinates that can vary separately. The rest of the paper studies what that count forces once one asks for exact resolution.

2.2 Degrees of Freedom

Definition 2.2 (Degrees of Freedom). The quantity $\text{DOF}(A) \in \mathbb{N}$ counts independent coordinates of variation in a bounded decision system A . In the mechanized development it is the local structural parameter attached to **Architecture**; later sections identify it exactly with the structural rank of a canonical decision problem.

Operational meaning. If $\text{DOF}(A) = n$, the system has n independent coordinates that must be resolved in the worst case by any exact resolver.

¹ Lean: [L45](#)

Proposition 2.3 (DOF Additivity). *For disjoint bounded decision systems A_1 and A_2 :*

$$\text{DOF}(A_1 \oplus A_2) = \text{DOF}(A_1) + \text{DOF}(A_2)$$

2

Proof. Independent coordinate sets combine by disjoint union, so the coordinate count is additive under composition. ■

2.3 Finite Physical Acquisition

Theorem 2.4 (Counting Gap). *Let $\varepsilon, C \in \mathbb{N}$ with $\varepsilon > 0$ and $C > 0$. If each information-acquisition event consumes ε discrete cost units, then*

$$\varepsilon \cdot N \leq C \implies N \leq C.$$

Equivalently, any bounded system with positive per-event cost admits only finitely many acquisition events.

3

Proof. In $\mathbb{N}$ every positive integer is at least one, so $N = 1 \cdot N \leq \varepsilon N \leq C$. ■

Bounded capacity plus positive per-event cost already forbids infinite checking. Physics enters when the abstract cost unit is calibrated. Under Landauer-type calibration, exact irreversible acquisition carries positive cost, so bounded material systems inherit the counting gap directly.

Proposition 2.5 (Bounded Region). *A bounded physical region is characterized by diameter $d > 0$ and signal speed $c > 0$. Its maximum information-acquisition rate is c/d events per unit time.*

4

Theorem 2.6 (Bounded Acquisition Rate). *For a bounded region with diameter d , signal speed c , and operating time T ,*

$$\text{acquisitions}(T) \leq \frac{cT}{d}.$$

In particular, acquisition count is finite on finite horizons.

5

Proof. Signals require at least d/c time to traverse the region, so no more than c/d acquisition events can occur per unit time. ■

Theorem 2.7 (Discrete Acquisition). *Information acquisition is realized by discrete state transitions. A bounded physical decision process is therefore a finite transition system whose acquisition events are countable transition points.*

6

Theorem 2.8 (One Transition, One Bit). *Each elementary acquisition transition carries one boolean bit. Boolean coordinates are therefore the primitive units of exact physical information exchange in the canonical model.*

7

Together, Theorems 2.6, 2.7, and 2.8 identify the later binary decision encoding as the natural finite acquisition model. A bounded resolver acquires information through finitely many discrete events, and each such event contributes one elementary boolean distinction.

² : L17, L19 ³ : BA10 ⁴ : BA1 ⁵ : BA2 ⁶ : BA3 ⁷ : BA4

Theorem 2.9 (Resolution Requires a Sufficient Coordinate Set). *Any exact physical resolver for a decision problem must read a sufficient coordinate set. If fewer coordinates are read, there exist states indistinguishable to the resolver but requiring different optimal actions.*

8

Proof. If the accessed coordinates are not sufficient, two states agree on every read coordinate while disagreeing on the optimal action. Any resolver limited to those coordinates must therefore err on at least one of the two states. ■

2.4 Canonical Decision Encoding

Definition 2.10 (Canonical Decision Problem). For a bounded decision system A with $\text{DOF}(A) = n$, the canonical decision problem

$$\text{canonicalDP}(A)$$

has state space $\text{Fin } n \rightarrow \text{Bool}$ and action space $\text{Fin } n \oplus \text{Unit}$. Action $\text{inl}(i)$ queries coordinate i ; the fallback action $\text{inr}(\star)$ receives constant utility 1. Query action i receives utility 2 exactly when coordinate i is true and 0 otherwise.

The encoding is the exact Lean object `canonicalDP` in `Leverage/BridgeToDQ.lean`. Every coordinate is relevant by construction, so the encoding exposes the full interaction dimension of the source system.

The encoding records, in the smallest exact-resolution object, the coordinate structure that any bounded physical resolver must already confront. The Boolean coordinates represent primitive acquisition events.

2.5 Structural Rank

Definition 2.11 (Structural Rank). The *structural rank* of a finite decision problem is the cardinality of its relevant coordinate set, equivalently the size of any minimal sufficient set. It is the minimum interaction dimension that must be read to determine the optimal action exactly.

For the canonical decision problem attached to A , the relevant coordinate set is all of $\text{Fin } \text{DOF}(A)$, so the structural-rank problem is exactly matched to the local degree-of-freedom count.

2.6 Decision Quotient and Entropy

Definition 2.12 (Decision Quotient). For a decision problem D , states are identified when they induce the same optimal-action set. The quotient space of these equivalence classes is the *decision quotient* of D .

Proposition 2.13 (Optimizer Quotient as Canonical Exact Abstraction). *Let $\text{Opt} : S \rightarrow \mathcal{P}(A)$ be the optimizer map of a finite decision problem. In **Set**, the decision quotient is the coimage of Opt , canonically equivalent to $\text{im}(\text{Opt})$ [12]. Any surjective decision-preserving summary factors through this quotient.*

9

The quotient is the coarsest exact abstraction of the decision problem: it forgets only decision-irrelevant distinctions and preserves every distinction needed for exact action selection. The entropy and thermodynamic bounds below are stated for this canonical exact abstraction.

⁸ : BA5 ⁹ : QT1-3, QT5, QT7

Definition 2.14 (Decision Entropy). Let $\text{numOptClasses}(D)$ be the number of equivalence classes in the decision quotient. Two entropy normalizations are used:

$$H_{\text{bits}}(D) = \log_2 \text{numOptClasses}(D), \quad H_{\text{nats}}(D) = \log \text{numOptClasses}(D).$$

The physics results are naturally stated in nats because Landauer calibration contributes the factor $k_B T$ per nat of resolved decision information [10, 1].

2.7 Formalization in Lean

The local degree-of-freedom object lives in `Leverage/Foundations.lean`, while the canonical decision encoding and its rank-identification theorems live in `Leverage/BridgeToDQ.lean`. Structural rank and decision entropy are formalized in the decision-quotient development.

3 Structural Rank and Decision Entropy

The canonical decision encoding has two immediate consequences. The degree-of-freedom count is exactly the interaction dimension of the encoded decision problem, and exact physical resolution must pay for that interaction dimension in discrete bit events. Structural rank is the physical count of irreducible coordinate reads required by exact resolution.

3.1 Degree of Freedom Equals Structural Rank

Theorem 3.1 (DOF–Structural-Rank Identity). *For every bounded decision system A ,*

$$\text{srnk}(\text{canonicalDP}(A)) = \text{DOF}(A).$$

10

Proof. Write $n = \text{DOF}(A)$. By Definition 2.10, the state space is $\text{Fin } n \rightarrow \text{Bool}$, query action $\text{inl}(i)$ has utility 2 exactly when coordinate i is true and utility 0 otherwise, and the fallback action has utility 1. Fix any coordinate i and choose two states that agree everywhere except at i , with one state setting i to true and the other setting i to false; then $\text{inl}(i)$ is optimal in the first state and not optimal in the second, so erasing coordinate i changes the optimizer. Thus every coordinate in $\text{Fin } n$ is relevant, the relevant-coordinate set has cardinality n , and the structural rank is n . Substituting $n = \text{DOF}(A)$ gives the claim. ■

Corollary 3.2 (Rank-One Regime). *For every bounded decision system A ,*

$$\text{DOF}(A) = 1 \iff \text{srnk}(\text{canonicalDP}(A)) = 1.$$

11

Corollary 3.3 (Higher-Rank Regime). *For every bounded decision system A ,*

$$\text{DOF}(A) > 1 \implies \text{srnk}(\text{canonicalDP}(A)) > 1.$$

12

Theorem 3.4 (Minimum Physical Bit Operations). *Any exact resolver for $\text{canonicalDP}(A)$ requires at least $\text{DOF}(A)$ elementary bit-acquisition events.*

¹⁰ : L43 ¹¹ : L51 ¹² : L46

Proof. By Theorem 2.9, exact resolution requires reading a sufficient coordinate set. The structural-rank theorem implies every sufficient set has cardinality at least $\text{srnk}(\text{canonicalDP}(A))$. Theorem 3.1 identifies that rank with $\text{DOF}(A)$, so at least $\text{DOF}(A)$ coordinate reads are required. By Theorem 2.8, each read contributes one elementary bit-acquisition event. ■

3.2 Decision-Quotient Size

Theorem 3.5 (Decision-Class Bound). *For the canonical binary decision problem attached to a bounded decision system A ,*

$$\text{numOptClasses}(\text{canonicalDP}(A)) \leq 2^{\text{DOF}(A)}.$$

Proof. For binary coordinate spaces, the number of distinct optimal-action classes is at most 2^{srnk} . Apply that theorem to the canonical encoding and substitute 3.1. ■

Theorem 3.6 (Decision-Entropy Bound). *For the canonical binary decision problem attached to a bounded decision system A ,*

$$H_{\text{bits}}(\text{canonicalDP}(A)) \leq \text{DOF}(A), \quad H_{\text{nats}}(\text{canonicalDP}(A)) \leq \text{DOF}(A) \ln 2.$$

Proof. The bit-entropy statement is the entropy-rank inequality for binary coordinate spaces, again composed with Theorem 3.1. The nat-entropy statement is obtained by multiplying by $\ln 2$. ■

3.3 Formalization

The structural-rank bridge is formalized in `Leverage/BridgeToDQ.lean`; the minimum-bit and entropy bounds are formalized in the decision-quotient physics and information development. These are the objects used directly by the thermodynamic theorems of the next section.

4 Thermodynamic Consequences

The previous section identified degree of freedom with structural rank and bounded the entropy of the decision quotient. Exact resolution requires irreducible bit events. Matter pays for them.

4.1 Landauer-Linear Resolution Cost

Theorem 4.1 (Rank Controls Exact-Resolution Cost). *Let A be a bounded decision system and let M be a thermodynamic model with positive per-bit conversion constant. Then*

$$M.\text{joulesPerBit} \cdot \text{DOF}(A) \leq \text{energyLowerBound}(M, \text{DOF}(A)).$$

In particular, exact-resolution cost is at least linear in the degree-of-freedom count.

¹³ : BA5-6, L43 ¹⁴ : IT4, L43 ¹⁵ : IT3, L43 ¹⁶ : BA6-7, L43

Proof. Theorem 3.4 gives a lower bound of $\text{DOF}(A)$ elementary bit-acquisition events for exact resolution. The bounded-acquisition energy theorem then converts that bit lower bound into the displayed energy lower bound. ■

Theorem 4.2 (Rank-One Ground State). *If $\text{DOF}(A) = 1$, then every exact-resolution cycle for the canonical problem has energy at least one Landauer unit. If $\text{DOF}(A) > 1$, then the system lies strictly above that ground state.*

17

Proof. The rank-one statement is exactly the imported ground-state theorem for structural rank 1 (BA8). For the higher-rank regime, Corollary 3.3 gives $\text{srnk} > 1$, and Theorem 4.1 then places the resulting exact-resolution cycle strictly above the one-Landauer-unit floor. ■

4.2 Energy–Information Duality

Theorem 4.3 (Energy–Information Duality). *Let $D = \text{canonicalDP}(A)$, and let E be the realized energy for one exact-resolution cycle. If Landauer calibration holds at positive Boltzmann constant and temperature, then*

$$E \geq k_B T H_{\text{nats}}(D).$$

Equivalently, the minimum exact-resolution cost is at least $k_B T$ times the natural-log entropy of the decision quotient.

18

Proof. The entropy-rank inequality gives

$$H_{\text{nats}}(D) \leq \text{DOF}(A) \ln 2$$

by Theorem 3.6. Theorem 4.1 gives the complementary lower bound

$$E \geq \text{DOF}(A) k_B T \ln 2.$$

Comparing the two right-hand sides yields the announced inequality. In the mechanized artifact this is the energy-information theorem (EI1). ■

Corollary 4.4 (Unique Minimum-Cost Regime). *Among bounded decision systems in the canonical binary encoding, the unique minimum-cost regime is $\text{DOF}(A) = 1$.*

19

Proof. Theorem 4.2 identifies $\text{DOF}(A) = 1$ as the one-Landauer-unit ground state, while every bounded decision system with more than one degree of freedom lies strictly above it. ■

4.3 Worked Examples

Two toy canonical systems fix the scale of the bound.

One relevant coordinate. Take states $\{0, 1\}$ and actions $\{a_0, a_1\}$, with $\text{Opt}(0) = \{a_0\}$ and $\text{Opt}(1) = \{a_1\}$. Then $\text{DOF}(A) = 1$, the decision quotient has two classes, $H_{\text{nats}}(D) = \ln 2$, and Theorem 4.3 gives

$$E \geq k_B T \ln 2.$$

This is the rank-1 ground regime.

¹⁷ : BA8, L54–55 ¹⁸ : EI1, IT3, L43 ¹⁹ : BA8, L54–55

Two independent coordinates. Take states $\{00, 01, 10, 11\}$ and actions $\{a_{00}, a_{01}, a_{10}, a_{11}\}$, with each state having its matching optimal action. Then $\text{DOF}(A) = 2$, the decision quotient has four classes, $H_{\text{nats}}(D) = \ln 4 = 2 \ln 2$, and Theorem 4.3 gives

$$E \geq 2k_B T \ln 2.$$

Relative to the one-coordinate case, the minimum exact-resolution cost doubles.

4.4 Optimal-Transport Witness

The Landauer route lower-bounds exact resolution through irreversible bit acquisition. A complementary witness measures separation between future distributions on the integrity space $\{\text{intact}, \text{compromised}\}$. It does not replace the main proof chain, but it supplies an independent transport-theoretic signal that multiple distinguishable futures have nonzero cost.

Remark 4.5 (Optimal-Transport Bridge). The same separation admits an independent transport-cost witness on the two-state integrity space $\{\text{intact}, \text{compromised}\}$. The diagonal coupling has zero transport cost in the single-future regime (W1). Any coupling with off-diagonal mass has positive transport cost (W2). If the intact future mass dominates the compromised future mass, the intact state minimizes total transport to the future distribution (W3). If both future states carry positive mass, then transport from either pure state is strictly positive (W4). Multiple distinguishable futures therefore force positive transport cost independently of the Landauer route.

20

A transport witness and the Landauer witness emphasize different structures: one counts irreducible coordinate reads, while the other measures geometric separation of future mass. In the two-state integrity model they point in the same direction, since a single future has zero transport cost whereas genuinely split futures force strictly positive transport cost.

4.5 Interpretation

If degree of freedom is read as the number of independent physical coordinates that can vary separately, then lower DOF means lower exact-resolution cost because fewer independent coordinates must be resolved.

4.6 Formalization

The local bridge from degree of freedom to structural rank is formalized in `Leverage/BridgeToDQ.lean`. The physical acquisition and Landauer theorems are imported from the decision-quotient physics stack, in particular `Physics/BoundedAcquisition.lean` and `ThermodynamicLift.lean`. The role of the local `Architecture` object in this section is to provide the coordinate count transported into those theorems.

5 Convergence of Rank, Tractability, and Cost

Degree of freedom equals structural rank, and structural rank bounds decision entropy. The same rank-1 regime also appears as tractable sufficiency and minimum thermodynamic cost. An imported coherence theorem gives a separate single-source interpretation of that same point.

²⁰ : W1-4

5.1 Imported Coherence Reading

Theorem 5.1 (Coherent Single-Source Regime). *A bounded decision system lies in the coherent unit-independent-rate regime if and only if $\text{DOF}(A) = 1$.*

21

In the imported coherence development, rank 1 means exactly one locus is authoritative, every remaining encoding is a derived view, and all reachable states remain coherent.

5.2 Structural Rank

Theorem 5.2 (Rank Identification). *For every bounded decision system A ,*

$$\text{srnk}(\text{canonicalDP}(A)) = \text{DOF}(A).$$

22

The degree-of-freedom count is not separate from the decision-theoretic object; it is exactly the interaction dimension of the canonical decision problem.

5.3 Tractability Boundary

Theorem 5.3 (Tractable Sufficiency at Rank One). *In the canonical decision problem family, structural rank 1 is the tractable sufficiency regime, while higher structural rank enters the hard regime.*

23

When exactly one relevant coordinate survives, exact sufficiency certification is tractable. Once more than one relevant coordinate survives, the canonical family crosses into the imported hardness regime.

5.4 Thermodynamic Selection

Theorem 5.4 (Thermodynamic Selection). *In the canonical decision encoding, every bounded decision system with $\text{DOF}(A) > 1$ lies strictly above the rank-1 Landauer ground state in per-cycle resolution cost.*

24

The theorem uses only the rank identity together with Landauer calibration. Stronger hardness consequences require additional imported hypotheses.

5.5 Convergence Theorem

Theorem 5.5 (Convergence). *For every bounded decision system A , the following conditions are equivalent:*

$$\text{DOF}(A) = 1 \iff \text{srnk}(\text{canonicalDP}(A)) = 1 \iff \text{tractable sufficiency for canonicalDP}(A) \iff \text{minimum per}$$

25

²¹ : Imported coherence theorem (ORA1) ²² : L43, L46, L51 ²³ : Imported tractability theorem (L56, L53, L47)

²⁴ : L49, L54-55, BA8 ²⁵ : L43-44, L47, L55

The same rank-1 regime is simultaneously the exact one-coordinate regime, the tractable exact-certification regime, and the thermodynamic ground state. The imported coherence theorem gives an additional single-source interpretation of the same point.

Proof.

1. Theorem 5.2 identifies $\text{DOF}(A) = 1$ with structural rank 1.
2. Theorem 5.3 identifies structural rank 1 with the tractable sufficiency regime for the canonical family.
3. Theorem 5.4 identifies rank 1 as the unique minimum-cost thermodynamic regime.

Transitivity of logical equivalence completes the proof.

Theorem 5.1 supplies an imported single-source interpretation of the same rank-1 regime.

5.6 Formalization

The proof chain is explicit in the mechanized artifact. The local bridge theorems live in `Leverage/BridgeToDQ.lean`; the coherence theorem is imported from `Ssot`; and the tractability and Landauer-cost theorems are imported from the decision-quotient development. The proof provenance remains fully auditable.

5.7 England Replication Inequality

The England Replication Inequality is mechanized in `Leverage/BridgeToDQ.lean` (L45).

Theorem 5.6 (England Replication Inequality). *Let $\Delta S_{\min}(r) = r \cdot k_B \ln 2$ be the rank-indexed minimal entropy production under Landauer calibration. For the rank-1 ground regime and any replicated rank- k regime:*

$$\Delta S_{\min}(1) + k_B \ln k \leq \Delta S_{\min}(k)$$

equivalently, $\Delta S_{\min}(k) - \Delta S_{\min}(1) \geq k_B \ln k$.

²⁶

Proof. The gap is $(k - 1) \cdot k_B \ln 2$. Since $k \leq 2^{k-1}$ (L52), taking logs gives $\ln k \leq (k - 1) \ln 2$, so the gap is $\geq k_B \ln k$. ■

Modeling note. $\Delta S_{\min}$ is a definition within the model: the exact Landauer entropy cost of the canonical exact-resolution cycle. The “min” refers to physical optimality inside the calibrated decision model. In England’s 2013 stochastic-thermodynamic framework [6], the comparison point is the entropy premium associated with replication above a single-copy baseline. The present reformulation does not reproduce England’s full path-space dynamics or detailed-balance setting. It isolates the same $k_B \ln k$ multiplicity penalty as a finite-counting consequence of rank-indexed Landauer cost together with $k \leq 2^{k-1}$, making the selection penalty explicit in a finite exact-resolution model.

²⁶ : L45

5.8 Finite-Budget No-Collapse

Theorem 5.7 (Finite-Budget No-Collapse). *Let $B : \mathbb{N} \rightarrow \mathbb{N}$ be a budget profile, let $\text{ops} : \mathbb{N} \rightarrow \mathbb{N}$ be a required-operation profile, and let $\text{bitCost} > 0$ be the per-bit physical cost. If*

1. *B is globally bounded,*
2. *ops has an exponential lower bound, and*
3. *collapse means: for every input size n , some feasible bit budget realizes at least $\text{ops}(n)$ operations within budget $B(n)$,*

then no such physical collapse profile exists.

27

Proof. This is exactly the bounded-budget physical no-collapse theorem in the physical-hardness layer. Exponential growth eventually exceeds every fixed finite budget, and positive per-bit cost lifts that growth into an energy contradiction. ■

Remark 5.8 (Interpretation). Finite budget, positive event cost, and exponential exact-certification demand are jointly incompatible with a physical collapse model. Any stronger complexity-collapse conclusion requires an additional bridge from the chosen complexity claim to such a collapse profile.

Corollary 5.9 ($P = NP$ No-Go Transfer). *Assume a polynomial-collapse claim supplies a bridge to the finite-budget collapse profile of Theorem 5.7. Then that claim is physically impossible in the same model. In particular, any bridge from a $P = NP$ collapse claim to such a feasible physical-collapse profile yields a contradiction.*

28

Proof. Theorem 5.7 rules out the collapse profile itself. The transfer theorem states that if a $P = NP$ claim implies that profile, then the claim is false in the model. ■

6 Related Work

6.1 Landauer, Non-Equilibrium Thermodynamics, and Selection

Landauer’s principle gives the standard calibration from logically irreversible discrimination to minimum heat production and energy cost [10, 1]. Stochastic thermodynamics extends that floor to trajectory-level entropy production, work identities, and fluctuation relations [18, 22, 23, 8, 2]. Finite-time erasure and mismatch corrections sharpen the same theme for controlled nonequilibrium protocols [5, 17, 13]. The theorem chain above isolates a different object: a finite exact-resolution lower bound indexed by the number of independent coordinates that must be resolved to preserve the optimizer.

Relative to the Seifert and Van den Broeck–Esposito framework, the present model does not attempt a full trajectory description of a driven Markov process, housekeeping heat, or protocol-dependent dissipation. It gives instead a mechanized non-asymptotic lower bound in terms of structural rank and decision-quotient entropy. Conversely, stochastic-thermodynamic frameworks resolve time-dependent nonequilibrium refinements that are outside the current finite exact-resolution model.

27 : PH26 28 : PH14–15, PH26

Non-equilibrium selection and replication arguments lie in the same neighborhood [6]. The entropy premium used above is deliberately discrete: multiplicity is reduced to counting over a finite quotient family and the elementary inequality $k \leq 2^{k-1}$, rather than to a separate stochastic-process ansatz.

6.2 Zero-Error, Functional, and Quotient Information

The information object is closer to zero-error and confusability-based information theory than to average-case source coding [19, 9, 11, 3]. The central quantity is not full state entropy but the entropy of the decision quotient: how many distinct optimal-action classes survive after irrelevant coordinates are erased.

Function-relative information in physics and origins-of-life work also conditions information on successful function or selection [20, 24]. The object studied here is narrower and exact: coordinate erasure is admissible precisely when optimal-action correspondence is preserved. The rank-1 regime is therefore the one-coordinate exact-decision regime, the tractable sufficiency regime, and the minimum calibrated-cost regime.

6.3 Categorical Quotients and Exact Abstraction

Quotienting states by equality of Opt is the standard coimage construction for the decision quotient of the optimizer map $\text{Opt} : S \rightarrow \mathcal{P}(A)$ in **Set**, canonically equivalent to its image [12]. The novelty is not the existence of that quotient, but the theorem chain tying it to coordinate sufficiency, structural rank, decision entropy, and thermodynamic cost in one proof object.

6.4 Formal Verification and Machine-Checked Theory

The proofs are machine-checked in Lean 4 [4] against the Mathlib library [21]. Related mechanized precedents include verified computability and semantics developments in Coq and Isabelle [7, 16, 15] and certificate-carrying proof artifacts [14]. Mechanization matters because the main identifications are exact rather than rhetorical: degree of freedom, structural rank, quotient entropy, tractable sufficiency, and calibrated thermodynamic cost are kept distinct and then linked by explicit theorems.

7 Conclusion

7.1 Summary

The central result is the convergence theorem (Theorem 5.5): the rank-1 regime of the canonical decision encoding is simultaneously the one-coordinate regime, the tractable sufficiency regime, and the thermodynamic ground state. An imported coherence theorem gives a separate single-source reading of the same point.

Framework	Rank-1 means	Formal source
Local system parameter	$\text{DOF}(A) = 1$	Leverage/Foundations
Structural information	Structural rank = 1	Leverage/BridgeToDQ
Computational complexity	Tractable sufficiency checking	DecisionQuotient
Statistical physics	Minimum thermodynamic cost per cycle	BoundedAcquisition, ThermodynamicLi

The theorem package is a mathematical-physics statement: DOF is identified with structural rank, structural rank bounds decision entropy, and Landauer calibration turns that rank bound into an energy bound. All of these statements are machine-checked in Lean 4 with an auditable proof trail.

Main consequences:

- The exact identification of degree of freedom with structural rank in the canonical decision encoding.
- The energy-information theorem $E \geq k_B T H_{\text{nats}}(D)$ for exact-resolution cost.
- The unconditional thermodynamic selection statement that every higher-rank regime lies above the rank-1 Landauer ground state.
- The finite-budget no-collapse theorem: bounded budget, positive per-bit cost, and exponential lower-bound growth cannot coexist with physical collapse.
- The England replication inequality (L45): $\Delta S_{\min}(k) - \Delta S_{\min}(1) \geq k_B \ln k$, proved by finite counting rather than by an informal thermodynamic analogy.

7.2 Mechanized Status

The central thermodynamic and convergence statements are all theorem-level results in the mechanized artifact:

- L55: unconditional energy separation above the rank-1 ground state.
- L49: quantitative Landauer-linear energy bound.
- PH26: bounded budget plus positive bit-cost plus exponential lower bound implies no physical collapse.
- L45: $\Delta S_{\min}(k) - \Delta S_{\min}(1) \geq k_B \ln k$ (Theorem 5.6).

7.3 Limitations

1. Canonical-encoding scope: the main theorems are exact for the canonical binary decision encoding attached to the bounded decision system. Extending the same conclusions to more general physical encodings requires an explicit transport argument.

2. Imported hardness regime: the thermodynamic cost theorems are unconditional, but the stronger no-polynomial-certification claims rely on imported hardness results for the canonical family.

3. Calibration choice: Landauer calibration supplies the physical conversion constant. Stronger substrate-dependent lower bounds are possible, but they belong to a different modeling layer than the one studied here.

4. Finite-budget model class: the no-collapse theorem is a statement about globally bounded budget profiles with positive per-bit cost and exponential lower-bound growth. Different collapse claims require explicit bridges into that profile language.

7.4 Impact

For mathematical physics: the England Replication Inequality (Theorem 5.6) is reduced to finite counting plus Landauer calibration. The result gives a mechanized, non-asymptotic entropy gap for replication without appealing to an informal thermodynamic analogy.

For information theory and complexity: structural rank appears as the information coordinate of DOF, and the minimum-cost thermodynamic regime coincides with the tractable sufficiency regime.

For formalized theory building: the argument shows that physically meaningful theorem packages can be built from exact finite objects and kept fully auditable at the proof-artifact level.

7.5 Final Remarks

Finite thermodynamic and structural consequences of exact resolution follow from first principles. In the formal bounded-system-to-**canonical**DP map, the regime $\text{DOF} = 1$ is the unique point at which structural rank, tractable sufficiency, and minimum calibrated thermodynamic cost coincide. The imported coherence theorem supplies a separate single-source interpretation of that same point.

All theorems are machine-checked. The thermodynamic lower bound is unconditional, and the replication inequality gives the entropy premium for multiplicity as $k_B \ln k$ per cycle. A companion manuscript develops the software-engineering and case-study implications. The mathematical-physics core is isolated here.

A Lean Proof Artifacts

This appendix reports proof traceability directly from source and generated mapping artifacts.

A.1 Claim Coverage Matrix

Paper claim	Lean handle
Corollary 4.4: Unique Minimum-Cost Regime	BA8 , L54 , L55
Corollary 5.9: $P = NP$ No-Go Transfer	PH26 , PH15 , PH14
Corollary 3.3: Higher-Rank Regime	L46 , L51
Corollary 3.2: Rank-One Regime	L51
Proposition 2.5: Bounded Region	BA1
Definition 2.3: Bounded Decision System	L17 , L19
Definition 2.13: Canonical Decision Problem	QT2 , QT7 , QT1 , QT3
Theorem 2.6: Bounded Acquisition Rate	BA1 , BA2
Theorem 5.1: Coherent Single-Source Regime	ORA1
Theorem 2.4: Counting Gap	BA10
Theorem 2.7: Discrete Acquisition	BA3
Theorem 3.1: DOF–Structural-Rank Identity	L43
Theorem 4.3: Energy–Information Duality	IT3 , EI1 , L43
Theorem 4.1: Rank Controls Exact-Resolution Cost	BA7 , BA6 , L43
Theorem 5.6: England Replication Inequality	L45

Paper claim	Lean handle
Theorem 3.6: Decision-Entropy Bound	IT3 , L43
Theorem 5.7: Finite-Budget No-Collapse	PH26
Theorem 5.5: Convergence	L43 , L44 , L47 , L55
Theorem 3.4: Minimum Physical Bit Operations	BA5 , BA6 , L43 , L46
Theorem 3.5: Decision-Class Bound	IT4 , L43
Theorem 2.8: One Transition, One Bit	BA3 , BA4
Theorem 5.2: Rank Identification	L43 , L46 , L51
Theorem 4.2: Rank-One Ground State	BA8 , L54 , L55
Theorem 2.9: Resolution Requires a Sufficient Coordinate Set	BA5
Theorem 5.4: Thermodynamic Selection	BA8 , L49 , L54 , L55
Theorem 5.3: Tractable Sufficiency at Rank One	L47 , L53 , L56

Auto summary: mapped 26/26 (full=26, derived=0, unmapped=0).

A.2 Lean Handle Map

ID	Lean Handle / Source	ID	Lean Handle / Source
BA1	Physics.BoundedAcquisition.BoundedRegion paper4/DecisionQuotient/Physics/ BoundedAcquisition.lean	BA2	Physics.BoundedAcquisition.acquisition_rate_ bound paper4/DecisionQuotient/Physics/ BoundedAcquisition.lean
BA3	Physics.BoundedAcquisition.acquisitions_are_ transitions paper4/DecisionQuotient/Physics/ BoundedAcquisition.lean	BA4	Physics.BoundedAcquisition.one_bit_per_ transition paper4/DecisionQuotient/Physics/ BoundedAcquisition.lean
BA5	Physics.BoundedAcquisition.resolution_reads_ sufficient paper4/DecisionQuotient/Physics/ BoundedAcquisition.lean	BA6	Physics.BoundedAcquisition.srank_le_ resolution_bits paper4/DecisionQuotient/Physics/ BoundedAcquisition.lean
BA7	Physics.BoundedAcquisition.energy_ge_srank_ cost paper4/DecisionQuotient/Physics/ BoundedAcquisition.lean	BA8	Physics.BoundedAcquisition.srank_one_energy_ minimum paper4/DecisionQuotient/Physics/ BoundedAcquisition.lean
BA10	Physics.BoundedAcquisition.counting_gap_ theorem paper4/DecisionQuotient/Physics/ BoundedAcquisition.lean	EI1	ThermodynamicLift.energy_ge_kbt_nat_entropy paper4/DecisionQuotient/ThermodynamicLift.lean
IT3	DecisionQuotient.quotientEntropy_le_srank_ binary paper4/DecisionQuotient/Information.lean	IT4	DecisionQuotient.numOptClasses_le_pow_srank_ binary paper4/DecisionQuotient/Information.lean
L17	Leverage.compose_dof Leverage/Foundations.lean	L19	Leverage.composition_dof_additive Leverage/Theorems.lean
L43	dof_eq_srank Leverage/BridgeToDQ.lean	L44	dof_one_iff_max_leverage Leverage/Foundations.lean
L45	england_replication_inequality Leverage/BridgeToDQ.lean	L46	incoherent_srank_gt_one Leverage/BridgeToDQ.lean
L47	max_coherence_forces_tractability Leverage/BridgeToDQ.lean	L49	srank_energy_lower_bound Leverage/BridgeToDQ.lean
L51	ssot_srank_one Leverage/BridgeToDQ.lean	L52	succ_le_two_pow Leverage/BridgeToDQ.lean
L53	sufficiency_conp_hard	L54	thermodynamic_selection Leverage/BridgeToDQ.lean

(continued...)

ID	Lean Handle / Source	ID	Lean Handle / Source
L55	thermodynamic_selection_unconditional Leverage/BridgeToDQ.lean	L56	tractable_bounded_core paper4/DecisionQuotient/ClaimClosure.lean
ORA1	oracle_arbitrary paper2/Ssot/Coherence.lean	PH14	PhysicalComplexity.p_eq_np_physically_ impossible_of_collapse_map paper4/DecisionQuotient/Physics/ PhysicalHardness.lean
PH15	PhysicalComplexity.p_eq_np_physically_ impossible_canonical paper4/DecisionQuotient/Physics/ PhysicalHardness.lean	PH26	PhysicalComplexity.no_collapse_of_bounded_ budget_pos_cost_exp_lb paper4/DecisionQuotient/Physics/ PhysicalHardness.lean
QT1	DecisionProblem.quotient_is_coarsest paper4/DecisionQuotient/Quotient.lean	QT2	DecisionProblem.quotientMap_preservesOpt paper4/DecisionQuotient/Quotient.lean
QT3	DecisionProblem.quotient_represents_opt_equiv paper4/DecisionQuotient/Quotient.lean	QT7	DecisionProblem.quotient_has_unique_ factorization paper4/DecisionQuotient/Quotient.lean
W1	Physics.single_future_zero_cost paper4/DecisionQuotient/Physics/ WassersteinIntegrity.lean	W2	Physics.transportCost_pos_of_offDiag paper4/DecisionQuotient/Physics/ WassersteinIntegrity.lean
W3	Physics.integrity_is_centroid paper4/DecisionQuotient/Physics/ WassersteinIntegrity.lean	W4	Physics.wasserstein_bridge paper4/DecisionQuotient/Physics/ WassersteinIntegrity.lean

A.3 Proof Hardness Index

Paper handle	Hardness profile	Regime tags	Lean support
cor:	unspecified	-	BA8, L54, L55
minimum-cost-regime			
cor:pnp-nogo	unspecified	-	PH26, PH15, PH14
cor:rank-above-one	unspecified	-	L46, L51
cor:rank-one	unspecified	-	L51
prop:bounded-region	unspecified	-	BA1
prop:dof-additive	unspecified	-	L17, L19
prop:	unspecified	-	QT2, QT7, QT1, QT3
optimizer-quotient			
thm:	unspecified	-	BA1, BA2
bounded-acquisition			
thm:coherent-single-source	unspecified	-	ORA1
thm:counting-gap	unspecified	-	BA10
thm:	unspecified	-	BA3
discrete-acquisition			
thm:dof-srank	unspecified	-	L43
thm:energy-entropy	unspecified	-	IT3, EI1, L43
thm:energy-rank	unspecified	-	BA7, BA6, L43
thm:england	unspecified	-	L45
thm:entropy-bound	unspecified	-	IT3, L43
thm:finite-budget-no-collapse	unspecified	-	PH26
thm:five-way	unspecified	-	L43, L44, L47, L55
thm:	unspecified	-	BA5, BA6, L43, L46
min-bit-operations			
thm:numopt-bound	unspecified	-	IT4, L43
thm:one-transition-one-bit	unspecified	-	BA3, BA4
thm:	unspecified	-	L43, L46, L51
rank-identification			
thm:rank-one-ground	unspecified	-	BA8, L54, L55
thm:resolution-sufficient	unspecified	-	BA5
thm:thermodynamic-selection	unspecified	-	BA8, L49, L54, L55

Auto summary: indexed 26 claims by hardness profile (unspecified=26).

B Notes on assumptions and extensions

This appendix lists the principal modeling assumptions and common extensions relevant for the finite decision-thermodynamic framework of the paper:

- **Canonical encoding:** The main theorems are exact for the canonical binary decision problem attached to the bounded decision system. Other physical encodings require an explicit transport theorem.
- **Landauer calibration:** Thermodynamic cost is calibrated by a per-bit Landauer floor. Stronger substrate-dependent lower bounds may exist, but they are additional assumptions, not part of the theorem package under discussion.
- **Exact decision setting:** The results concern exact sufficiency and exact-resolution cost. Approximate, stochastic, or bounded-confidence regimes require separate analysis.
- **Finite state family:** The entropy and replication theorems are finite counting results. Continuum models must first be reduced to a finite decision quotient before these arguments apply.
- **Future work:** Natural extensions include richer transport theorems from physical encodings to canonical decision problems, continuous-information analogues, and stronger substrate-specific dissipation bounds.

C Complete Theorem Index

Paper-level labeled claims:

Foundations (Section 2):

- Proposition 2.3 (DOF Additivity)
- Theorem 2.4
- Proposition 2.5
- Theorem 2.6
- Theorem 2.7
- Theorem 2.8
- Theorem 2.9

Structural Rank and Decision Entropy (Section 3):

- Theorem 3.1

- Corollary [3.2](#)
- Corollary [3.3](#)
- Theorem [3.4](#)
- Theorem [3.5](#)
- Theorem [3.6](#)

Thermodynamic Consequences (Section 4):

- Theorem [4.1](#)
- Theorem [4.2](#)
- Theorem [4.3](#)
- Corollary [4.4](#)

Convergence (Section 5):

- Theorem [5.5](#)
- Theorem [5.6](#)
- Theorem [5.7](#)
- Corollary [5.9](#)

Primary Lean sources:

- `Leverage/Foundations.lean`
- `Leverage/BridgeToDQ.lean`
- `LambdaDR.lean`
- `Leverage.lean`

References

- [1] Charles H. Bennett. The thermodynamics of computation—a review. *International Journal of Theoretical Physics*, 21(12):905–940, 1982.
- [2] Gavin E. Crooks. Entropy production fluctuation theorem and the nonequilibrium work relation for free energy differences. *Physical Review E*, 60(3):2721–2726, 1999.
- [3] Imre Csiszár and János Körner. *Information Theory: Coding Theorems for Discrete Memoryless Systems*. Cambridge University Press, 2nd edition, 2011.
- [4] Leonardo de Moura and Sebastian Ullrich. The lean 4 theorem prover and programming language. In *International Conference on Automated Deduction*, pages 625–635. Springer, 2021.
- [5] Giovanni Diana, G. Baris Bagci, and Massimiliano Esposito. Finite-time erasing of information stored in fermionic bits. *Physical Review E*, 87(1):012111, 2013.

- [6] Jeremy L. England. Statistical physics of self-replication. *The Journal of Chemical Physics*, 139(12):121923, 2013.
- [7] Yannick Forster, Fabian Kunze, and Maxi Wuttke. Verified programming of Turing machines in Coq. In *Proceedings of the 9th ACM SIGPLAN International Conference on Certified Programs and Proofs (CPP 2020)*, pages 114–128. ACM, 2020.
- [8] C. Jarzynski. Nonequilibrium equality for free energy differences. *Physical Review Letters*, 78(14):2690–2693, 1997.
- [9] János Körner. Coding of an information source having ambiguous alphabet and the entropy of graphs. In *Transactions of the 6th Prague Conference on Information Theory, Statistical Decision Functions, Random Processes*, pages 411–425, 1973.
- [10] Rolf Landauer. Irreversibility and heat generation in the computing process. *IBM Journal of Research and Development*, 5(3):183–191, 1961.
- [11] László Lovász. On the shannon capacity of a graph. *IEEE Transactions on Information Theory*, 25(1):1–7, 1979.
- [12] Saunders Mac Lane. *Categories for the Working Mathematician*, volume 5 of *Graduate Texts in Mathematics*. Springer, 2 edition, 1998.
- [13] Pau Manzano, Uğur Kardeş, Édgar Roldán, and David H. Wolpert. Absolute bounds on the intrinsic dissipation of classical and quantum processes due to irreversibility and mismatch. *Physical Review X*, 14(2):021026, 2024.
- [14] George C. Necula. Proof-carrying code. In *Proceedings of the 24th ACM SIGPLAN-SIGACT Symposium on Principles of Programming Languages*, pages 106–119. ACM, 1997.
- [15] Tobias Nipkow and Gerwin Klein. *Concrete Semantics with Isabelle/HOL*. Springer, 2014.
- [16] Tobias Nipkow, Lawrence C. Paulson, and Markus Wenzel. *Isabelle/HOL: A Proof Assistant for Higher-Order Logic*. Springer, 2002.
- [17] Karel Proesmans, Jannik Ehrich, and John Bechhoefer. Finite-time landauer principle. *Physical Review Letters*, 125:100602, 2020.
- [18] Udo Seifert. Stochastic thermodynamics, fluctuation theorems and molecular machines. *Reports on Progress in Physics*, 75(12):126001, 2012.
- [19] Claude E. Shannon. Zero-error capacity of a noisy channel. *IRE Transactions on Information Theory*, 2(3):8–19, 1956.
- [20] Jack W. Szostak. Functional information: Molecular messages. *Nature*, 423(6941):689, 2003.
- [21] The mathlib Community. The Lean mathematical library. In *Proceedings of the 9th ACM SIGPLAN International Conference on Certified Programs and Proofs (CPP 2020)*, pages 367–381. Association for Computing Machinery, 2020.
- [22] Christian Van den Broeck and Massimiliano Esposito. Ensemble and trajectory thermodynamics: A brief introduction. *Physica A: Statistical Mechanics and its Applications*, 418:6–16, 2015.

- [23] David H. Wolpert. The stochastic thermodynamics of computation. *Journal of Physics A: Mathematical and Theoretical*, 52(19):193001, 2019.
- [24] Michael Wong, Carol Cleland, Daniel R. Arend, Stuart Bartlett, H. James Cleaves, Heather Demarest, Aditi Prabhu, Jonathan I. Lunine, and Robert M. Hazen. On the roles of function and selection in evolving systems. *Proceedings of the National Academy of Sciences*, 120(44):e2310223120, 2023.